

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} M_{xx} & M_{xy} & 0 \\ M_{xy} & -M_{xx} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{xx} = M_{dv}^{(1)}$$

$$M_{xy} = M_{dxy}^{(1)}$$

* Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzz} & -M_{xxz} & 0 \\ M_{xxz} & M_{yzz} & 0 \\ 0 & 0 & M_{yzz} \end{bmatrix}$$

$$M_{xxz} = M_{fbz}^{(1)}$$

$$M_{yzz} = M_{f3}^{(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxz} \\ 0 & 0 & -M_{xxz} \\ 0 & 0 & 0 \\ M_{yzz} & \frac{M_{xxz}}{2} & 0 \\ -\frac{M_{xxz}}{2} & M_{yzz} & 0 \\ 0 & 0 & -2M_{yzz} \end{bmatrix}$$

$$M_{xxz} = 2T_{dxy}^{(1)}$$

$$M_{yzz} = T_{dv}^{(1)}$$

$$\frac{M_{xxz}}{2} = T_{dxy}^{(1)}$$

$$2M_{yzz} = 2T_{dv}^{(1)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} M_{yzz} & M_{yzy} & 0 \\ M_{yzy} & -M_{yzz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = T_{dv}^{(2)}$$

$$M_{yzy} = T_{dxy}^{(2)}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} M_{xxxx} & 0 & M_{xxzz} & 0 & 0 & M_{xxxy} \\ 0 & -M_{xxxx} & -M_{xxzz} & 0 & 0 & M_{xxxy} \\ M_{xxzz} & -M_{xxzz} & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xxxy} & M_{xxxy} & M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxxx} = 2M_{dv}^{(1)} + 4M_{dv}^{(2)} + M_{gv}^{(1)}$$

$$M_{xxzz} = M_{dv}^{(1)} - M_{gv}^{(1)}$$

$$M_{xxxy} = M_{dxy}^{(1)} + 2M_{dxy}^{(2)} - M_{gbz}^{(1)}$$

$$M_{zzxy} = M_{dxy}^{(1)} + 2M_{gbz}^{(1)}$$

$$M_{yzyz} = -M_{dv}^{(2)} + M_{gv}^{(1)}$$

$$M_{yzzx} = M_{dxy}^{(2)} + 2M_{gbz}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} \\ -M_{xxyy} & 0 & -M_{xxzz} & 0 & 0 & M_{xxxy} \\ -M_{xxzz} & M_{xxzz} & 0 & 0 & 0 & M_{zzxy} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxxy} & -M_{xxxy} & -M_{zzxy} & 0 & 0 & 0 \end{bmatrix}$$

$$M_{xxyy} = -2M_{dv}^{(3)} + 4T_{f3}^{(1)}$$

$$M_{xxzz} = -M_{dv}^{(3)} - 4T_{f3}^{(1)}$$

$$M_{xxxy} = M_{dxy}^{(3)} + 2T_{fbz}^{(1)}$$

$$M_{zzxy} = M_{dxy}^{(3)} - 4T_{fbz}^{(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} M_{yzyz} & M_{yzzx} & 0 \\ M_{yzzx} & -M_{yzyz} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzyz} = 2M_{dv}^{(4)}$$

$$M_{yzzx} = 2M_{dxy}^{(4)}$$

* Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & 0 & M_{xxxy} \\ 0 & 0 & -M_{xxxy} \\ 0 & 0 & 0 \\ M_{yzyz} & M_{yzzx} & 0 \\ -M_{yzzx} & M_{yzyz} & 0 \\ 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{xxxy} = -2M_{dxy}^{(5)} + T_{fbz}^{(2)}$$

$$M_{yzyz} = -M_{dv}^{(5)} + T_{f3}^{(2)}$$

$$M_{yzzx} = -M_{dxy}^{(5)} - T_{fbz}^{(2)}$$

$$M_{xyxy} = 2M_{dv}^{(5)} + T_{f3}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & M_{yzyz} & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$M_{yzyz} = -M_{dv}^{(6)} + T_{f3}^{(3)}$$

$$M_{yzzx} = M_{dxy}^{(6)} + T_{fbz}^{(3)}$$

$$M_{xyxx} = -2M_{dxy}^{(6)} + T_{fbz}^{(3)}$$

$$M_{xyxy} = 2M_{dv}^{(6)} + T_{f3}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} M_{xxxx} & M_{xxyy} & M_{xxzz} & 0 & 0 & M_{xxxy} & 0 & 0 & M_{xxyx} \\ -M_{xxyy} & -M_{xxxx} & -M_{xxzz} & 0 & 0 & M_{xxyx} & 0 & 0 & M_{xxxxy} \\ M_{zzxx} & -M_{zzxx} & 0 & 0 & 0 & M_{zzxy} & 0 & 0 & M_{zzzxy} \\ 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzz} & M_{yzxz} & 0 \\ 0 & 0 & 0 & M_{yzzx} & -M_{yzzz} & 0 & M_{yzzx} & -M_{yzyz} & 0 \\ M_{xyxx} & M_{xyxx} & M_{xyzx} & 0 & 0 & M_{xyxy} & 0 & 0 & -M_{xyxy} \end{bmatrix}$$

$$\begin{aligned} M_{xxxx} &= 2M_{dv}^{(1)} + 4M_{dv}^{(2)} + M_{gv}^{(1)} \\ M_{xxyy} &= -2M_{dv}^{(3)} + 4T_{f3}^{(1)} \\ M_{xxzz} &= M_{dv}^{(1)} - M_{dv}^{(3)} - M_{gv}^{(1)} - 4T_{f3}^{(1)} \\ M_{xxxy} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} - 2M_{dxy}^{(5)} - M_{gbz}^{(1)} + 2T_{fbz}^{(1)} + T_{fbz}^{(2)} \\ M_{xxyx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} + 2M_{dxy}^{(5)} - M_{gbz}^{(1)} + 2T_{fbz}^{(1)} - T_{fbz}^{(2)} \\ M_{zzxx} &= M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gv}^{(1)} + 4T_{f3}^{(1)} \\ M_{zzxy} &= M_{dxy}^{(1)} + M_{dxy}^{(3)} + 2M_{gbz}^{(1)} - 4T_{fbz}^{(1)} \\ M_{yzyz} &= -M_{dv}^{(2)} - M_{dv}^{(5)} + M_{gv}^{(1)} + T_{f3}^{(2)} \\ M_{yzzx} &= M_{dxy}^{(2)} - M_{dxy}^{(5)} + 2M_{gbz}^{(1)} - T_{fbz}^{(2)} \\ M_{yzzz} &= -M_{dv}^{(2)} + M_{dv}^{(5)} + M_{gv}^{(1)} - T_{f3}^{(2)} \\ M_{yzxz} &= M_{dxy}^{(2)} + M_{dxy}^{(5)} + 2M_{gbz}^{(1)} + T_{fbz}^{(2)} \\ M_{xyxx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} - M_{dxy}^{(3)} - M_{gbz}^{(1)} - 2T_{fbz}^{(1)} \\ M_{xyzx} &= M_{dxy}^{(1)} - M_{dxy}^{(3)} + 2M_{gbz}^{(1)} + 4T_{fbz}^{(1)} \\ M_{xyxy} &= 2M_{dv}^{(5)} + T_{f3}^{(2)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & M_{yzyz} & M_{yzzx} & 0 & M_{yzzz} & M_{yzzx} & 0 \\ 0 & 0 & 0 & -M_{yzzx} & M_{yzzz} & 0 & -M_{yzzx} & M_{yzyz} & 0 \\ M_{xyxx} & -M_{xyxx} & 0 & 0 & 0 & M_{xyxy} & 0 & 0 & M_{xyxy} \end{bmatrix}$$

$$\begin{aligned} M_{yzyz} &= 2M_{dv}^{(4)} - M_{dv}^{(6)} + T_{f3}^{(3)} \\ M_{yzzx} &= 2M_{dxy}^{(4)} + M_{dxy}^{(6)} + T_{fbz}^{(3)} \\ M_{yzzz} &= -2M_{dv}^{(4)} - M_{dv}^{(6)} + T_{f3}^{(3)} \\ M_{yzxz} &= -2M_{dxy}^{(4)} + M_{dxy}^{(6)} + T_{fbz}^{(3)} \\ M_{xyxx} &= -2M_{dxy}^{(6)} + T_{fbz}^{(3)} \\ M_{xyxy} &= 2M_{dv}^{(6)} + T_{f3}^{(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} M_{xxxx} & M_{xxxx} & 0 \\ M_{xxxx} & -M_{xxxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & M_{yyzz} \\ M_{zzxx} & M_{zzxy} & 0 \\ M_{xxyx} & -M_{yyzz} - M_{zzxx} & 0 \\ M_{yzzx} & M_{yzzz} & 0 \\ 0 & 0 & M_{zzxx} \\ -M_{yzzz} - M_{zzxx} & M_{xyyy} & 0 \\ 0 & 0 & -\frac{M_{xxxy}}{2} - \frac{M_{xxyx}}{2} + M_{xyyy} + M_{zzxy} \end{bmatrix}$$

$$\begin{aligned}
M_{xxxx} &= 2M_{dv}^{(1)} + 4M_{dv}^{(2)} + M_{gv}^{(1)} \\
M_{xxx y} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} - 2M_{dxy}^{(5)} - M_{gbz}^{(1)} \\
M_{yyzz} &= -M_{dv}^{(1)} + M_{dv}^{(3)} + M_{gv}^{(1)} \\
M_{zzxx} &= M_{dv}^{(1)} + M_{dv}^{(3)} - M_{gv}^{(1)} \\
M_{zzxy} &= M_{dxy}^{(1)} + M_{dxy}^{(3)} + 2M_{gbz}^{(1)} \\
M_{xxyx} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} + M_{dxy}^{(3)} + 2M_{dxy}^{(5)} - M_{gbz}^{(1)} \\
M_{yzzx} &= M_{dxy}^{(2)} - M_{dxy}^{(5)} + 2M_{gbz}^{(1)} \\
M_{yzyy} &= -M_{dv}^{(2)} + M_{dv}^{(5)} + M_{gv}^{(1)} \\
M_{zzxz} &= M_{dv}^{(2)} + M_{dv}^{(5)} - M_{gv}^{(1)} \\
M_{xyyy} &= M_{dxy}^{(1)} + 2M_{dxy}^{(2)} - M_{dxy}^{(3)} - M_{gbz}^{(1)}
\end{aligned}$$